

Gaussian Frequency Shift Keying (GFSK)

A working note on the mathematics of the modulation index h and the bandwidth–time product BT

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1. What GFSK is, in one paragraph

GFSK sends bits by shifting a carrier’s frequency up or down. Before the bit stream reaches the modulator it passes through a Gaussian low-pass filter, which replaces the abrupt frequency steps of plain FSK with smooth S-curves. This narrows the transmitted spectrum dramatically at the cost of deliberate intersymbol interference. Because the information lives entirely in the phase, the envelope is constant, which is the property that makes GFSK radios cheap: the power amplifier can run saturated.

GFSK belongs to the family of **continuous phase modulation** (CPM) schemes. Everything below follows from writing it in that framework.

2. Signal model

2.1 The transmitted passband signal

The transmitted signal is a real, bandpass waveform:

$$s(t) = A \cos(2\pi f_c t + \phi(t; \mathbf{b}) + \phi_0), \quad A = \sqrt{\frac{2E_b}{T}}$$

Symbol	Meaning	Units
A	carrier amplitude, constant	V

Symbol	Meaning	Units
E_b	energy per bit	J
T	symbol (bit) period, $T = 1/R_b$	s
f_c	carrier frequency	Hz
$\phi(t; \mathbf{b})$	information-bearing phase, defined in §2.3	rad
ϕ_0	unknown constant carrier phase	rad
$\mathbf{b} = \{b_n\}$	data sequence, $b_n \in \{-1, +1\}$	—

Why that amplitude. The instantaneous power of $A \cos(\cdot)$ averages to $A^2/2$, so the energy delivered in one symbol period is $A^2 T/2 = E_b$. Choosing $A = \sqrt{2E_b/T}$ is what makes E_b mean “energy per bit” and lets §3.3 quote error rates in terms of E_b/N_0 without extra bookkeeping.

2.2 Passing to complex baseband

Carrying f_c (a few GHz) through every derivation is pointless when all the information sits within a megahertz of it. The standard device is the **complex lowpass equivalent**, and it deserves an explicit definition rather than an assertion.

Given a real bandpass $s(t)$, form the analytic signal using the Hilbert transform $\hat{s}(t)$, then de-rotate by the carrier:

$$s_+(t) = s(t) + j\hat{s}(t), \quad \tilde{s}(t) \triangleq s_+(t) e^{-j2\pi f_c t}$$

The inverse — the relation you actually use — is

$$s(t) = \text{Re} \{ \tilde{s}(t) e^{j2\pi f_c t} \}$$

This is exact provided $s(t)$ is **narrowband**: its occupied bandwidth must be much less than f_c , so that the positive- and negative-frequency images don’t overlap. For GFSK that condition is satisfied by orders of magnitude — Bluetooth occupies about 1 MHz at $f_c \approx 2.4$ GHz, a ratio of 4×10^{-4} .

Applying it here is just Euler’s formula:

$$s(t) = A \cos(2\pi f_c t + \phi(t; \mathbf{b}) + \phi_0) = \text{Re} \left\{ \underbrace{A e^{j(\phi(t; \mathbf{b}) + \phi_0)}}_{\tilde{s}(t; \mathbf{b})} e^{j2\pi f_c t} \right\}$$

$$\tilde{s}(t; \mathbf{b}) = A e^{j(\phi(t; \mathbf{b}) + \phi_0)}$$

Two simplifications, stated rather than smuggled in:

1. **ϕ_0 is dropped.** It is a fixed rotation of the entire constellation, absorbed into the receiver’s phase reference. It matters only when discussing carrier recovery (§7), where it reappears as the quantity being estimated.
2. **A is normalized to 1.** Amplitude is a scale factor with no effect on pulse shape, spectrum, h , or BT . Setting $A = 1$ gives the unit-energy form $\tilde{s}(t; \mathbf{b}) = e^{j\phi(t; \mathbf{b})}$ used from §3 onward. When a signal-to-noise calculation is needed, restore $A = \sqrt{2E_b/T}$.

So the constant-envelope claim, stated precisely: $|\tilde{s}(t; \mathbf{b})| = A$ for all t **and for every** $\mathbf{b}$ — the data moves the phase around the unit circle and never off it. **The envelope is exactly constant, not approximately so** — it falls out of the model rather than being engineered in, and it is the entire reason GFSK survives in battery-powered devices.

Convention warning. Texts differ on where factors of $\sqrt{2}$ and $\frac{1}{2}$ live in the bandpass–baseband map. Some define $s(t) = \sqrt{2} \operatorname{Re}\{\tilde{s}e^{j2\pi f_c t}\}$ so that $\tilde{s}$ carries the same *power* as s ; some use $\frac{1}{2}$ or work with $2 \operatorname{Re}\{\cdot\}$. All are self-consistent and all give the same h , BT , and pulse shapes — but they disagree on how noise variance maps between domains. Fix your convention before writing a link budget. This note uses $s = \operatorname{Re}\{\tilde{s}e^{j2\pi f_c t}\}$ throughout.

2.3 The information-bearing phase

$$\phi(t; \mathbf{b}) = 2\pi h \sum_{n=-\infty}^{\infty} b_n q(t - nT)$$

On the notation. The semicolon separates the continuous variable t from the parameter $\mathbf{b}$ that indexes which waveform we are talking about. It is worth carrying rather than abbreviating to $\phi(t)$, because the sum runs over *all* n — every symbol, past and future, contributes to the phase at every instant. That is the defining property of continuous phase modulation and the source of everything awkward and everything useful about it:

- The modulator has memory, so ϕ at time t is not recoverable from b_n alone (§5).
- The set of reachable values of ϕ is finite for rational h , which is what makes the trellis (§3.2).
- Detection is *sequence* estimation: the receiver picks $\hat{\mathbf{b}} = \arg \max_{\mathbf{b}} \Lambda(r | \mathbf{b})$, searching over candidate sequences rather than deciding symbol by symbol (§7.1). Without the $\mathbf{b}$ in the notation there is nothing visible to maximize over.

Where a single fixed sequence is under discussion and no confusion can arise, $\phi(t)$ is used as shorthand.

$q(t)$ is the **phase pulse**, the running integral of the **frequency pulse** $g(t)$:

$$q(t) = \int_{-\infty}^t g(\tau) d\tau$$

with the normalization

$$\int_{-\infty}^{\infty} g(\tau) d\tau = \frac{1}{2} \quad \Longleftrightarrow \quad q(+\infty) = \frac{1}{2}$$

This normalization is a convention. With it, each transmitted symbol contributes exactly

$$\Delta\phi_n = 2\pi h \cdot b_n \cdot \frac{1}{2} = \pi h b_n$$

to the accumulated phase.

2.4 Why the area is 1/2 and not 1

Nothing forces the choice, and it is worth seeing why. Let the area be a free constant

$$\kappa \triangleq \int_{-\infty}^{\infty} g(\tau) d\tau$$

Repeating the derivations of §2.3 and §3.1 with κ left symbolic:

$$\Delta\phi_n = 2\pi h \kappa b_n, \quad \Delta f = \frac{h\kappa}{T}, \quad h = \frac{\Delta f T}{\kappa}$$

Only the product $h\kappa$ is physically observable. The pairs $(h, \kappa) = (0.32, \frac{1}{2})$ and $(0.16, 1)$ generate bit-for-bit identical waveforms with identical spectra. Choosing κ does not constrain the physics; it only decides how the factor of two is split between the pulse and the index.

So why $\kappa = 1/2$? Because h is older than the CPM formalism. It descends from the analog FM **deviation ratio** $\beta = \Delta f / f_m$, with f_m the highest modulating frequency. A binary stream at rate R_b has its highest fundamental at $R_b/2$ — the alternating ... 1010 ... pattern is a square wave at half the bit rate — giving

$$h = \frac{\Delta f}{R_b/2} = \frac{2\Delta f}{R_b}$$

which was standard long before anyone wrote FSK as a CPM. The $1/2$ is chosen precisely so the CPM machinery reproduces this pre-existing definition.

The payoff is that the landmark values of h come out at memorable numbers. Coherent FSK tones are orthogonal over T when their separation is a multiple of $1/2T$; the minimum case $2\Delta f = 1/2T$ gives $h = 1/2$ exactly, with phase advancing $\pm\pi/2$ per symbol — *minimum* shift keying, in quadrature, which is what connects it to offset-QPSK in §7.2.

Landmark	$\kappa = \frac{1}{2}$ (this note, and standard)	$\kappa = 1$
Phase advance per symbol	$\pi h b_n$	$2\pi h b_n$
Coherent orthogonality (MSK)	$h = 1/2$	$h = 1/4$
Non-coherent orthogonality (Sunde's)	$h = 1$	$h = 1/2$
Optimal binary CPFSK distance	$h \approx 0.715$	$h \approx 0.357$
Relation to deviation	$h = 2\Delta f T$	$h = \Delta f T$

Both columns are internally consistent. The right one is simply a relabeling in which every quoted value of h halves and the name “minimum shift keying” stops pointing at anything.

Practical consequence. When a datasheet or paper quotes an h that looks off by a factor of two, check κ before assuming an error. This is the third convention trap in this note, alongside the 3 dB definition (§4.1) and the bandpass–baseband scaling (§2.2).

2.5 Instantaneous frequency

Differentiating,

$$f_{\text{dev}}(t; \mathbf{b}) = \frac{1}{2\pi} \frac{\partial \phi(t; \mathbf{b})}{\partial t} = h \sum_n b_n g(t - nT)$$

This is the quantity plotted in a frequency-versus-time trace. For plain FSK it is a square wave; for GFSK it is the rounded version.

3. The modulation index h

3.1 Definition and its relation to peak deviation

Send an infinite run of $b_n = +1$. Then

$$f_{\text{dev}} = h \sum_n g(t - nT) = h \cdot \frac{1}{T} \int_{-\infty}^{\infty} g(\tau) d\tau = \frac{h}{2T}$$

(the sum of a pulse's shifted copies converges to its area divided by the spacing, regardless of pulse shape). Define this steady-state value as the **peak frequency deviation** Δf :

$$\Delta f = \frac{h}{2T} \iff h = 2\Delta f T = \frac{2\Delta f}{R_b}$$

Two readings of the same equation:

- h is the **tone separation** $2\Delta f$ measured in units of the symbol rate.
- h is the **phase swing per symbol** divided by π .

Note that $g(t)$'s *shape* never entered this derivation, only its area. **h and BT are therefore independent knobs:** h sets how far the phase travels, BT sets how smoothly it gets there.

3.2 The phase tree and trellis

Because phase accumulates, the modulator has memory. Sampling at symbol boundaries, the accumulated phase after m symbols is

$$\theta_m = \pi h \sum_{n=1}^m b_n \pmod{2\pi}$$

(written θ rather than ϕ to keep the symbol-rate state distinct from the continuous-time phase, and indexed by m because N is reserved for samples per symbol in §8.1).

If h is rational, $h = k/p$ in lowest terms, θ_m can only ever take p distinct values modulo 2π (for even k) or $2p$ values (for odd k). This finite set is the **phase state**. Combined with the $L - 1$ most recent symbols still inside the pulse (§4.4), it forms a complete state description: the pair $(\theta_m, b_{m-1}, \dots, b_{m-L+1})$ determines the waveform on the next symbol interval. That is what collapses the infinite sequence dependence in $\phi(t; \mathbf{b})$ into a finite trellis, and it is why maximum-likelihood *sequence* detection is tractable via the Viterbi algorithm rather than requiring a search over all $2^{|\mathbf{b}|}$ candidates. Irrational h gives an unbounded state space and is never used deliberately.

3.3 Values of h worth knowing

h	Name / property
0.5	MSK . Minimum h for which the two tones are orthogonal over T under coherent detection. Phase advances exactly $\pm\pi/2$ per symbol.
≈ 0.715	Maximizes minimum Euclidean distance for binary CPFSK: $d_{\min}^2 \approx 2.43$ versus 2.0 for MSK, a gain of about 0.85 dB.

h	Name / property
1.0	Sunde's FSK. Minimum h for orthogonality under <i>non-coherent</i> detection. Spectrally wasteful; discrete carrier spikes appear.

Distance translates to error rate through

$$P_b \approx Q\left(\sqrt{d_{\min}^2 \frac{E_b}{N_0}}\right)$$

with $d_{\min}^2 = 2$ recovering the familiar BPSK/QPSK curve.

3.4 Why real systems tolerate a range of h

Bluetooth BR specifies h anywhere in $[0.28, 0.35]$. This is not sloppiness. Many cheap transmitters modulate a VCO or PLL open-loop, and the deviation depends on analog gain that drifts with temperature, supply, and process. The receiver is designed to cope. A consequence worth remembering: **the receiver cannot assume it knows h exactly**, which pushes designs toward non-coherent or differential detection that degrade gracefully with h error.

4. The Gaussian filter and the BT product

4.1 The filter

The Gaussian is most transparent in its own natural parameterization, as a width in frequency:

$$H_G(f) = \exp\left(-\frac{f^2}{2\sigma_f^2}\right)$$

The mapping from the engineering specification (B , a 3 dB bandwidth in Hz) to the time-domain width σ_t then splits into two independent steps. Keeping them apart matters, because the two constants that appear have unrelated origins and respond to different convention choices.

Step 1: the 3 dB point fixes σ_f . Impose $|H_G(B)|^2 = \frac{1}{2}$:

$$\exp\left(-\frac{B^2}{\sigma_f^2}\right) = \frac{1}{2} \Rightarrow \boxed{\sigma_f = \frac{B}{\sqrt{\ln 2}}}$$

This is the *only* place $\sqrt{\ln 2} \approx 0.8326$ enters. It is a consequence of choosing the half-power point as the definition of bandwidth and nothing else. Substituting back gives the equivalent closed form

$$H_G(f) = \exp\left(-\frac{\ln 2}{2} \left(\frac{f}{B}\right)^2\right)$$

Step 2: Gaussian time–frequency reciprocity fixes σ_t . The Fourier transform of a Gaussian is a Gaussian, and their width parameters are exactly reciprocal:

$$\sigma_t \sigma_f = \frac{1}{2\pi}$$

Composing the two steps:

$$\sigma_t = \frac{1}{2\pi\sigma_f} = \frac{\sqrt{\ln 2}}{2\pi B} \approx \frac{0.1325}{B}$$

giving the unit-area impulse response

$$h_G(t) = \frac{1}{\sqrt{2\pi}\sigma_t} \exp\left(-\frac{t^2}{2\sigma_t^2}\right)$$

Where the 2π comes from. It is purely an artifact of measuring bandwidth in hertz. The Fourier kernel is $e^{j2\pi ft}$ when f is in cycles per second, so that factor has to surface somewhere in any time–frequency width product expressed in Hz. In angular frequency it vanishes outright:

$$\sigma_t \sigma_\omega = 1, \quad \omega = 2\pi f$$

That is the “natural” form. We nonetheless quote B in Hz because BT is only dimensionless that way, and BT is the parameter every standard specifies.

The same filter, four ways. The 2π is *conserved*: it can be relocated but not removed. Suppress it in the transform step and it reappears in the filter definition. The four parameterizations below all describe the identical filter:

Parameter	Filter written as	Relation to the others	Units
B	$\exp\left(-\frac{\ln 2}{2}(f/B)^2\right)$	the specification	Hz
σ_f	$\exp\left(-f^2/2\sigma_f^2\right)$	$\sigma_f = B/\sqrt{\ln 2}$	Hz
α	$\exp\left(-\alpha f^2\right)$	$\alpha = 1/2\sigma_f^2 = \ln 2/2B^2 = 2\pi^2\sigma_t^2$	s ²
σ_t	$\exp\left(-2\pi^2\sigma_t^2 f^2\right)$	$\sigma_t = 1/2\pi\sigma_f = \sqrt{\ln 2}/2\pi B$	s

Read the last row against the third: parameterizing by the time-domain width puts the 2π **in the definition**, after which the inverse transform is free of it. Parameterizing by B , σ_f , or α keeps the definition clean and pays the 2π **at the crossing**. Note also that α is not a “pure” frequency-domain quantity despite appearing in one — it has units of s² and equals $2\pi^2\sigma_t^2$, so it silently carries the factor squared.

In angular frequency the accounting disappears entirely, which is the real content of the claim above:

$$H_G(\omega) = \exp\left(-\frac{\sigma_t^2 \omega^2}{2}\right) \leftrightarrow h_G(t) \propto \exp\left(-\frac{t^2}{2\sigma_t^2}\right), \quad \sigma_\omega = \frac{1}{\sigma_t}$$

Two Gaussians with reciprocal widths and no loose constants anywhere.

Why this is not pedantry. The FIR filter in §8.1 is built in *samples*, so what the code actually needs is

$$\sigma_n = \sigma_t f_s = \frac{\sqrt{\ln 2}}{2\pi} \cdot \frac{N}{BT} \approx \frac{0.1325 N}{BT}$$

For $BT = 0.5$ and $N = 8$ samples/symbol this is $\sigma_n \approx 2.12$ samples. Dropping or double-counting the 2π here misplaces the filter width by a factor of 6.28 — not a subtle error, but one that produces a filter which still *runs*, still looks Gaussian on a plot, and simply transmits the wrong spectrum.

So the two constants in $\sigma_t = \sqrt{\ln 2}/2\pi B$ are doing separate jobs, and it is worth knowing which is which:

Constant	Origin	Changes if...
$\sqrt{\ln 2}$	the 3 dB (half-power) bandwidth definition	you reference a different attenuation level, or define B on $ H $ instead of $ H ^2$
2π	frequency measured in Hz, not rad/s	you switch to angular frequency, where it disappears

Convention trap: some texts define B at the 3 dB point of $|H|$ rather than $|H|^2$, which replaces $\ln 2$ with $2 \ln 2$ and rescales σ_f by $\sqrt{2}$. Check before comparing numbers across references.

A note on the uncertainty principle. The relation $\sigma_t \sigma_f = 1/2\pi$ above is reciprocity between the *amplitude* Gaussians $h_G(t)$ and $H_G(f)$. The Heisenberg–Gabor uncertainty principle is a statement about the *energy densities* $|h_G(t)|^2$ and $|H_G(f)|^2$, whose widths are each smaller by $\sqrt{2}$. Writing those as $\tilde{\sigma}_t = \sigma_t/\sqrt{2}$ and $\tilde{\sigma}_\omega = \sigma_\omega/\sqrt{2}$:

$$\tilde{\sigma}_t \tilde{\sigma}_\omega = \frac{\sigma_t \sigma_\omega}{2} = \frac{1}{2}$$

which meets the bound $\tilde{\sigma}_t \tilde{\sigma}_\omega \geq 1/2$ with equality — Gaussians are the unique minimizers. Same underlying fact as Step 2, a factor of two apart. This is why no pulse shape can beat the Gaussian’s simultaneous concentration in time and frequency, and it is the deep reason §4.3’s tradeoff between spectral width and ISI has no free lunch in it.

From here on, σ without a subscript means σ_t , the time-domain width.

4.2 The resulting frequency pulse

Start from the rectangular CPFSK pulse, centred:

$$g_{\text{REC}}(t) = \begin{cases} \frac{1}{2T}, & |t| \leq T/2 \\ 0, & \text{otherwise} \end{cases}$$

and convolve:

$$g(t) = (g_{\text{REC}} * h_G)(t) = \frac{1}{2T} \int_{-T/2}^{T/2} h_G(t - \tau) d\tau$$

The integral of a Gaussian over an interval is a difference of Gaussian CDFs, so in closed form:

$$g(t) = \frac{1}{2T} \left[Q\left(\frac{2\pi B}{\sqrt{\ln 2}} \left(t - \frac{T}{2}\right)\right) - Q\left(\frac{2\pi B}{\sqrt{\ln 2}} \left(t + \frac{T}{2}\right)\right) \right]$$

where $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du$. Convolution preserves area, so $\int g = 1/2$ still holds and §3.1 is untouched.

4.3 Why only BT matters

Rewrite in normalized time $u = t/T$:

$$\frac{\sigma}{T} = \frac{\sqrt{\ln 2}}{2\pi BT} \approx \frac{0.1325}{BT}$$

Every property of the pulse shape — its width in symbols, the amount of ISI, the spectral rolloff — depends on B and T **only through the product BT** . This is the whole reason the parameter is quoted as a product rather than as an absolute filter bandwidth.

BT	σ/T	Pulse span (99% energy)	Character
∞	0	1 symbol	rectangular CPFSK (no filtering)
1.0	0.13	≈ 1.5 symbols	mild smoothing
0.5	0.265	≈ 2 symbols	Bluetooth / BLE
0.3	0.442	≈ 3 symbols	GSM
0.2	0.663	≈ 4 symbols	very narrow, heavy ISI

Lower $BT \rightarrow$ wider pulse $\rightarrow$ narrower spectrum $\rightarrow$ more ISI. There is no free lunch anywhere on this axis.

4.4 Truncation

The Gaussian has infinite support, so implementations truncate $g(t)$ to L symbol periods and renormalize the area back to $1/2$. Typical choices are $L = 3$ for $BT = 0.5$ and $L = 4$ for $BT = 0.3$. L is the **CPM memory length** and it propagates directly into receiver complexity (§6.1).

5. Intersymbol interference: the price of small BT

When $g(t)$ spans $L > 1$ symbols, the frequency at any instant is a weighted blend of L consecutive bits. Two practical consequences:

Reduced effective deviation. An isolated bit surrounded by opposite bits never reaches the full $\pm\Delta f$; the phase trajectory turns around before it gets there. The eye closes. Long runs of identical bits do reach full deviation, so the frequency trace of real data shows *data-dependent* peak deviation — which is exactly why deviation specs are measured on defined test patterns (Bluetooth uses 00001111 for max deviation and 10101010 for the alternating case).

A larger trellis. For $h = k/p$ and alphabet size M , the number of trellis states is

$$N_{\text{states}} = p \cdot M^{L-1}$$

GMSK at $h = 1/2$, $M = 2$, $L = 4$ gives $2 \cdot 2^3 = 16$ states. The ISI is *known and deterministic*, so a Viterbi detector recovers nearly all of the loss — GMSK at $BT = 0.3$ costs roughly 0.5 dB in E_b/N_0 relative to unfiltered MSK.

6. Spectrum

6.1 Shape

The exact CPM power spectral density has no simple closed form, because $\phi(t; \mathbf{b})$ depends non-linearly on $\mathbf{b}$. But the qualitative result is clear from the pulse:

- Rectangular pulses have sinc-shaped transforms, so sidelobes decay as $1/f^2$ in power — a polynomial tail.
- The Gaussian factor $\exp(-\frac{\ln 2}{2}(f/B)^2)$ decays faster than **any** polynomial.

That difference is what turns adjacent-channel splatter from a problem into a non-problem.

6.2 Carson's rule

A quick estimate of occupied bandwidth, treating the modulation as FM with highest baseband component $R_b/2$:

$$B_{\text{occ}} \approx 2 \left(\Delta f + \frac{R_b}{2} \right) = R_b (h + 1)$$

For Bluetooth ($h = 0.32$, $R_b = 1$ Mb/s) this gives 1.32 MHz. The true 99% bandwidth is closer to 1 MHz — Carson's rule is an *overestimate* for filtered CPM and should be used only for sizing, never for compliance.

6.3 Fractional power bandwidth

Bandwidth containing a given fraction of total power, normalized to R_b . These are indicative figures from the classic GMSK literature (Murota & Hirade) and vary with truncation length and pulse convention:

BT	90%	99%	99.9%
0.2	≈ 0.52	≈ 0.79	≈ 0.99
0.25	≈ 0.57	≈ 0.86	≈ 1.09
0.5	≈ 0.69	≈ 1.04	≈ 1.33
∞ (MSK)	≈ 0.78	≈ 1.20	≈ 2.76

Read the 99.9% column to see the point: going from MSK to $BT = 0.25$ shrinks the far-out spectrum by more than a factor of two.

7. Detection

7.1 Coherent, optimal

Maximum-likelihood sequence estimation: choose $\hat{\mathbf{b}} = \arg \min_{\mathbf{b}} \int |r(t) - \tilde{s}(t; \mathbf{b})|^2 dt$, evaluated efficiently by the Viterbi algorithm over the phase trellis of §3.2. Best performance, highest complexity, and it needs accurate carrier phase *and* accurate knowledge of h .

7.2 The Laurent decomposition

A result that makes GMSK receivers practical: any binary CPM signal with modulation index h can be written **exactly** as a sum of 2^{L-1} linearly amplitude-modulated pulse trains,

$$\tilde{s}(t; \mathbf{b}) = \sum_{k=0}^{2^{L-1}-1} \sum_n a_{k,n}(\mathbf{b}) C_k(t - nT)$$

For GMSK, the principal component C_0 carries roughly 99% of the signal energy. So you can treat GMSK as approximately *linear* — essentially offset-QPSK with a funny pulse — and use a single matched filter to C_0 instead of a full trellis, at a fraction of a dB. This approximation is why linear equalizers work on GSM.

7.3 Non-coherent options

- **Limiters–discriminator:** hard-limit, then differentiate the phase. Simple, robust to h uncertainty, no carrier recovery. Costs roughly 1–3 dB.
- **Differential detection:** correlate the received phase against itself delayed by T . Similar cost, very cheap in digital.

Most Bluetooth and sub-GHz ISM radios use one of these two, which is consistent with the loose h tolerance in §3.4.

8. Implementation

8.1 Digital baseband chain

1. Map bits to $b_n \in \{-1, +1\}$.
2. Upsample by N samples per symbol (insert $N - 1$ zeros, or repeat).
3. Filter with a truncated Gaussian FIR of span LN taps, normalized to unit sum $\rightarrow d[k] \in [-1, 1]$. The tap weights are $\exp(-n^2/2\sigma_n^2)$ with the width **in samples** given by §4.1:

$$\sigma_n = \sigma_t f_s = \frac{\sqrt{\ln 2}}{2\pi} \cdot \frac{N}{BT} \approx \frac{0.1325 N}{BT}$$

4. Accumulate phase.
5. Take sine and cosine.

The phase step per sample is

$$\phi[k] = \phi[k-1] + \frac{2\pi\Delta f}{f_s} d[k] = \phi[k-1] + \frac{\pi h}{N} d[k]$$

using $\Delta f = h/2T$ and $f_s = N/T$. Then

$$I[k] = \cos \phi[k], \quad Q[k] = \sin \phi[k]$$

A CORDIC rotator or a lookup table handles step 5; the accumulator must wrap modulo 2π cleanly.

8.2 Analog / PLL modulation

The alternative is to modulate a fractional-N synthesizer directly. The problem: the PLL loop filter is a low-pass that swallows the low-frequency content of the modulation, distorting long runs of identical bits. **Two-point modulation** solves this by injecting the data at two places — into the fractional divider (which the loop passes) and into the VCO tuning port (which the

loop rejects) — with the two paths' gains matched so the sum is flat. Gain mismatch is the usual cause of h error, hence §3.4.

9. Real systems

System	R_b	Δf	h	BT	Detection
Bluetooth BR	1 Mb/s	160 kHz nom.	0.32 (0.28–0.35)	0.5	non-coherent
Bluetooth LE 1M	1 Mb/s	250 kHz	0.5 (0.45–0.55)	0.5	non-coherent
GSM (GMSK)	270.833 kb/s	67.71 kHz	0.5 exactly	0.3	Viterbi + equalizer
DECT	1.152 Mb/s	≈ 288 kHz	0.5	0.5	non-coherent
Z-Wave	9.6–100 kb/s	varies	≈ 0.5 –1.0	0.5	non-coherent

Worked example: Bluetooth LE 1M

Given $R_b = 1$ Mb/s, $h = 0.5$, $BT = 0.5$.

$$T = 1 \mu\text{s}, \quad \Delta f = \frac{h}{2T} = \frac{0.5}{2 \times 10^{-6}} = 250 \text{ kHz}$$

Tone separation $2\Delta f = 500$ kHz. Gaussian 3 dB bandwidth:

$$B = \frac{BT}{T} = \frac{0.5}{1 \mu\text{s}} = 500 \text{ kHz}, \quad \sigma = \frac{\sqrt{\ln 2}}{2\pi B} = \frac{0.8326}{2\pi \times 5 \times 10^5} \approx 265 \text{ ns}$$

which is $0.265T$, matching the table in §4.3. Phase advance per bit is $\pi h = \pi/2$. Carson's estimate: $R_b(h+1) = 1.5$ MHz against a 2 MHz channel spacing — comfortable, and the real occupied bandwidth is well under that.

For a modulator at $f_s = 8$ MHz, $N = 8$ samples/symbol, phase step $= \pi h/N = \pi/16$ radians per sample at full deviation, and a Gaussian FIR of $L \times N = 3 \times 8 = 24$ taps.

10. Choosing the parameters

Pick h first. It is set by the link budget and the receiver you can afford.

- Coherent receiver, spectrum at a premium $\rightarrow h = 0.5$ (MSK/GMSK), the sweet spot for orthogonality and trellis simplicity.
- Non-coherent receiver, sensitivity matters more than bandwidth $\rightarrow h$ toward 0.7–1.0.
- Low h (below ≈ 0.3) shrinks bandwidth but the tones crowd together and sensitivity suffers.

Then pick BT against the channel mask and the receiver's ISI tolerance.

- $BT \geq 0.5$: ISI is mild enough for a simple discriminator. Default for anything cost-sensitive.
- $BT \approx 0.3$: needs an equalizer or Viterbi detector, buys real spectral efficiency. Justified when channels are tightly packed.
- $BT < 0.25$: rarely worth it. ISI grows faster than the spectral gain.

Sanity checks before committing:

- $\Delta f = h/2T$ must fit the channel spacing with margin.
- $\sigma/T = 0.1325/BT$ tells you the pulse span; confirm L covers it.
- $p \cdot M^{L-1}$ is your trellis size; confirm the receiver can afford it.
- Verify which 3 dB convention your filter reference uses (§4.1).

Appendix: symbols

Symbol	Meaning	Units
$s(t)$	real transmitted passband signal	V
$\tilde{s}(t; \mathbf{b})$	complex lowpass equivalent, $s = \text{Re}\{\tilde{s}e^{j2\pi f_c t}\}$	V
A	carrier amplitude, $\sqrt{2E_b/T}$	V
f_c	carrier frequency	Hz
ϕ_0	unknown constant carrier phase	rad
$\phi(t; \mathbf{b})$	information-bearing phase; semicolon marks dependence on the whole sequence	rad
θ_m	accumulated phase state after m symbols, mod 2π	rad
E_b	energy per bit	J
b_n	data symbol, ± 1	—
T	symbol period	s
$R_b = 1/T$	bit rate	bit/s
h	modulation index	—
Δf	peak frequency deviation	Hz
B	Gaussian filter 3 dB bandwidth	Hz
BT	bandwidth–time product	—
$\sigma \equiv \sigma_t$	Gaussian impulse response width	s
σ_f	Gaussian filter width in frequency, $= B/\sqrt{\ln 2}$	Hz
σ_n	Gaussian FIR width in samples, $= \sigma_t f_s$	samples
α	filter exponent coefficient, $= 2\pi^2 \sigma_t^2$	s ²
$g(t)$	frequency pulse, area 1/2	1/s
$q(t)$	phase pulse, $q(\infty) = 1/2$	—
κ	frequency pulse area, $= 1/2$ here (see §2.4)	—
L	pulse truncation length	symbols
N	samples per symbol	—
$d_{\min}^2$	normalized min. squared Euclidean distance	—

Key relations

$$h = 2\Delta f T \quad \Delta f = \frac{h}{2T} \quad \Delta\phi_n = \pi h b_n$$

$$\sigma = \frac{\sqrt{\ln 2}}{2\pi B} \quad \frac{\sigma}{T} = \frac{0.1325}{BT} \quad N_{\text{states}} = p M^{L-1}$$

Further reading

- Anderson, Aulin & Sundberg, *Digital Phase Modulation* — the definitive CPM treatment.
- Murota & Hirade, “GMSK Modulation for Digital Mobile Radio Telephony,” *IEEE Trans. Comm.*, 1981 — the original GMSK spectral analysis.
- Laurent, “Exact and Approximate Construction of Digital Phase Modulations by Superposition of AM Pulses,” *IEEE Trans. Comm.*, 1986.